

# RUPAM SEN

RTL Design Engineer - Vervesemi Microelectronics

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## PROFESSIONAL SUMMARY

ASIC Design Engineer with 2.5+ years of experience in RTL design and digital IP development. Owning the crucial digital blocks of Vervesemi's first motor control MCU project and participated in two successful tape-out. Skilled in Verilog/SystemVerilog, micro-architecture definition, implementation of motor-control IPs and standard communication protocols. Strong expertise in Verification, Synthesis and Gate Level Simulation. Strong understanding of LINT, CDC, Assertion, Coverage and UPF. Adept at cross-team collaboration and design bring-up.

## WORK EXPERIENCE

- VERVESEMI MICROELECTRONICS - Digital Design Engineer II** Oct, 2023 - Present
  - MCPWM - MOTOR CONTROL PULSE WIDTH MODULATION**
    - Designed the RTL for the **MCPWM module** to generate upto 12 independent PWM outputs, developed core features including dead-time generation, fault detection, and output override control.
    - Performed **IP-level verification** using SystemVerilog testbenches over an **AXI-Lite interface** and developed **C-based environments** for SoC-level verification.
    - Created constraint file, and verified design functionality through post-synthesis simulation and timing analysis.
  - CORDIC COPROCESSOR**
    - Developed **CORDIC Coprocessor** to compute trigonometric and hyperbolic functions (sin, cos, tanh, arctan, square root, and modulus) using the **CORDIC algorithm and Digit-by-digit method**.
    - Achieved **up to 15-bit precision** on 16-bit outputs, and optimized pipeline architecture for **improved performance and area efficiency**.
    - Developed an **APB interface-based** environment for IP-level functional verification. Performed SoC-level verification and Gate-level simulation.
  - HALL SENSOR INTERFACE**
    - Designed the RTL for Hall sensor interface block to capture the BLDC motor's rotor position and speed.
    - Developed interpolation logic to generate a **high-resolution 16-bit rotor angle** from low-resolution Hall sensor inputs.
    - Provided support for **FPGA-based validation**. Prepared technical documentations and datasheet.
- CADENCE DESIGN SYSTEMS - Product Validation Intern** Aug, 2022 - July, 2023
  - Experienced on the **SV-AMS, Wreal, and RNM techniques**. Designed AMS testcases to validate **Xcelium**.
  - Conducted daily **regression analysis**. Identify the root cause of the failed regression testcases and debugging. Experienced on the version control system - **Perforce**.

## TECHNICAL SKILLS

|                |          |           |      |                 |
|----------------|----------|-----------|------|-----------------|
| Verilog        | C/C++    | Lint      | AXI  | Cadence Xcelium |
| System-Verilog | Python   | CDC       | APB  | Synopsis VCS    |
| Verilog-AMS    | Perl     | Assertion | UART | AMD Vivado      |
| UVM            | Makefile | Coverage  | I2C  | Cadence Genus   |
| UPF            | TCL      | Synthesis | SPI  |                 |

## EDUCATION

|                                                        |                                                                                        |            |             |
|--------------------------------------------------------|----------------------------------------------------------------------------------------|------------|-------------|
| <b>Master of Technology</b><br>[Communication Systems] | Motilal Nehru National Institute of Technology<br>(MNNIT), Allahabad                   | <b>9.1</b> | <b>2023</b> |
| <b>Bachelor of Technology</b><br>[ECE]                 | University of Calcutta – Institute of Radio Physics<br>and Electronics (IRPE), Kolkata | 8.76       | 2021        |